Econometrics II

Event Studies, DID and Rate Models

For today

1. Event studies
   a. Survey of case studies
   b. Dependence on ‘Market Model’

2. Alternatives to Event studies:
   a. Difference in Difference approach

3. Forecasting Events
   a. Poisson Rate Models
   b. Hazard Models

What is an event study

1. Event studies are popular in the corporate finance and accounting literature for testing the implications of corporate policy changes on **market efficiency**

2. Systematically abnormal security returns persisting after a particular class of corporate events are typically not consistent with market efficiency
   a. This is particularly true of long horizon event studies which are less subject to the impact of market microstructure issues (order imbalances, market maker inventory adjustments etc)

3. Separately from corporate finance, event studies have also been used in the field of law and economics to examine the effect of regulatory changes, as well as to assess the impact of legal case decisions

4. From both of the above examples, we can therefore organize existing event studies into the following 2 categories:
   a. Long horizon: "Before and After" event studies, which study the change in an outcome variable (e.g. security returns) before, and after a corporate (or otherwise) policy change or announcement
   b. Short horizon: In a short window (typically on the order of several days) around the policy change or announcement, what is the effect on the outcome variable, typically market excess or beta adjusted returns

5. From 1974 to 2000, the total number of papers in 5 academic journals ("Journal of Business", "Journal of Finance (JF)", "Journal of Financial Economics (JFE)", "Journal of Financial and Quantitative Analysis" and "Review of Financial Studies") utilizing an event study methodology was 565, with over 200 papers each in JF and JFE, with the basic statistical methodology not changing significantly during this 30 years

Survey of case studies

1. Departing from the format of our first 3 classes, we will overview case studies for corporate finance event studies first, before a critique of methodology

2. Following each case study, we will briefly summarize the learning points from any variations in their methodology, especially on the differences between short horizon and long horizon event studies

3. Events in the literature encompass both internally initiated changes in corporate policies, and also changes due to external factors

    a. Examples of internally initiated changes: (1) seasoned equity offering, (2) stock buybacks

    b. Examples of external factors: (3) bond rating downgrades, (4) inclusion on 'sustainability' indices

Case study 1: Seasoned / secondary equity offerings

Table 1

Yearly Common Stock Offering Announcements by NYSE/AMEX Listed Industrial Firms and Regulated Utilities

	Total	Industrial Offerings				Public Utilities Offerings		
Year	Offerings	Total	Primary	Combination*	Dual	Total	Primary	Dual
1963	4	2	1	0	1	2	1	1
1964	17	10	5	5	0	7	5	2
1965	11	11	5	6	0	0	0	0
1966	17	10	7	3	0	7	5	2
1967	27	24	10	13	1	3	3	0
1968	55	48	22	21	5	7	7	0
1969	51	38	18	19	1	13	11	2
1970	61	36	19	15	2	25	19	6
1971	154	109	44	58	7	45	25	20
1972	115	65	31	32	2	50	41	9
1973	62	16	12	4	0	46	41	5
1974	67	9	7	2	0	58	45	13
1975	117	33	22	8	3	84	59	24
1976	123	46	35	8	3	77	59	18
1977	91	20	4	6	10	71	60	11
1978	126	52	28	10	14	74	67	7
1979	110	44	25	10	9	66	63	3
1980	198	117	93	21	3	81	73	8
Total	1406	690	388	241	51	716	584	131

*Combination primary-secondary stock offerings were made entirely by industry firms with the exception of one pubic utility offering in 1975.

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=961034

1. Ronald W. Masulis, Ashok N. Korwar, **Seasoned equity offerings: An empirical investigation**, Journal of Financial Economics, Volume 15, Issues 1–2
2. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=961034
3. Authors study price adjustment of common stock to announcements of common stock offerings
4. Intuitively, a negative stock price change will be observed due to both signalling as well as anticipated supply-demand imbalances
5. Authors find that the negative stock price reaction is larger for industrial firms relative to public utilities **(what are hypotheses as to why?)**

Case study 1: Seasoned / secondary equity offerings

Table 2

Number of Equity Offerings per NYSE/AMEX Listed Stock
(1963-1980)

Number of Offerings per Firm	Industrials			Public Utilities		
	Firms	Offerings		Firms	Offerings	
	(No.)	(No.)	(%)	(No.)	(No.)	(%)
1	428	428	62	19	19	3
2	95	190	27	15	30	4
3	18	54	8	12	36	5
4	3	12	2	14	56	8
5	2	10	1	14	70	10
6	0	0	0	10	60	8
7	0	0	0	16	112	16
8	0	0	0	8	64	9
9	0	0	0	9	81	11
10	0	0	0	4	40	6
11	0	0	0	5	55	8
12	0	0	0	4	48	7
13	0	0	0	2	26	4
14	0	0	0	0	0	0
15	0	0	0	1	15	2
Total	546	694	100	133	712	101

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=961034

1. While secondary equity offerings attract both negative signalling as well as supply-demand implications, authors hypothesize that the negative implications will be stronger for industrials rather than public utilities

2. This is because public utilities are highly regulated not just in their service offerings, but also in their investment and also financing decisions.

3. Their stock offerings may require state utility commission approval

4. Hence, principal-agent conflicts suggested by a secondary equity offering may be less marked for utilities relative to industrial firms

5. On the 'principal-agent conflict' in SEOs:
   a. Who is the principal?
   b. Who is the agent?
   c. What are some hypothesized conflict(s)?
   d. What are some ways of testing hypotheses on such conflict of interests in the event study framework?

Case study 1: Event time returns

Table 5
Common Stock Portfolio Returns Surrounding Primary Offering Announcements of Seasoned Common Stock by Industrial Firms and Public Utilities†

	Industrial Firms (N=388)			Public Utilities (N=584)		
Event Day	Portfolio Returns (%)	Ave. Mkt. Returns (%)	Negative Stk Returns (%)	Portfolio Returns (%)	Ave. Mkt. Returns (%)	Negative Stk. Returns (%)
-20	.45	.16	37	.12	.14	35
-19	.39	.17	42	.10	.15	37
-18	.41	.18	44	.15	.18	35
-17	.19	.10	41	.06	.16	38
-16	.27	.17	44	.08	.14	34
-15	.14	.17	43	.11	.13	35
-14	.42	.19	38	.01	.16	40
-13	.40	.12	43	.05	.12	37
-12	.19	.10	41	.06	.09	38
-11	.38	.10	42	.08	.07	35
-10	.09	.10	43	-.05	.07	40
-9	.30	.18	43	.06	.10	36
-8	-.13	.11	48	.00	.14	38
-7	.16	.12	45	-.06	.13	39
-6	.51	.19	38	.06	.08	38
-5	.28	.18	40	.01	.07	39
-4	.11	.14	46	.03	.10	39
-3	.30	.10	40	-.03	.06	42
-2	.21	.17	42	-.07	.06	40
-1	-.08	.17	49	-.12	.07	41
0	-2.20	.05	71	-.39	.07	50
1	-1.05	.01	58	-.29	.05	47
2	.07	-.00	48	-.07	.02	40
3	.06	.06	44	-.11	.04	43
4	.15	.05	44	.04	.10	39
5	.11	.02	50	-.03	.10	40
6	-.14	.01	51	.01	.09	40
7	-.20	-.02	48	-.02	.06	38
8	.15	-.02	41	.09	.05	39
9	-.09	.02	46	.02	.10	39
10	.00	.01	45	.01	.08	38
11	.08	.01	45	.03	.11	36
12	.03	.05	45	.09	.05	35
13	.12	.03	41	.05	.06	37
14	.07	-.00	46	.02	.05	35
15	.35	.10	40	.03	.06	34
16	.02	-.03	48	.03	.07	36
17	-.05	-.04	46	.03	.07	34
18	.12	.02	43	.06	.03	32
19	-.06	-.03	45	.06	.04	34
20	.06	-.03	44	-.04	-.03	36
21	-.19	.00	50	.17	.03	32

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=961034

1. Table on the right shows daily abnormal returns for the portfolio of firms at a specific date ***relative*** to the buyback announcement

2. Given we are indexing to announcement date and not to implementation / execution date, do you think any 'abnormal' returns illustrate more strongly the effects of
   a. Informational signalling
   b. Supply-demand imbalances so induced
   c. Both of the above?

3. Plots in event time **prior** to SEO announcement can help to uncover corporate management incentives.

4. Based on pre versus post-announcement window returns for industrial firms, do you think there is evidence for either / both of the below:
   a. Superior managerial information on 'fair' valuation for the stock
   b. Moral hazard

5. What about speed of information diffusion and market efficiency?

Case study 1: Seasoned / secondary equity offerings: study of information diffusion

(industrials)	(utilities)
Pre-Announcement Period††:	
Portfolio Mean Daily Return =.31% (.161)	Portfolio Mean Daily Return = .05% (.066)
Market Mean Daily Return = .17% (.042)	Market Mean Daily Return = .11% (.039)
Percent of Stock Returns < 0 = 41.8 (2.67)	Percent of Stock Returns < 0 = 37.2 (1.94)
Post-Announcement Comparison Period††:	
Portfolio Mean Daily Return =.06% (.140)	Portfolio Mean Daily Return = .02% (.052)
Market Mean Daily Return = .04% (.050)	Market Mean Daily Return = .06% (.052)
Percent of Stock Returns < 0 = 44.7 (2.91)	Percent of Stock Returns < 0 = 36.0 (2.29)

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=961034

1. Authors also find SEOs where managers are able to reduce their shareholdings in the companies (by selling to external investors) result in incrementally more negative stock price reactions to the SEO announcement compared to SEOs where managers do not sell

2. This is consistent with both principal-agent conflict, and also a motivation of risk-shifting / moral hazard

Summarizing case study 1

- Formatting ‘abnormal’ returns into the pre and post announcement period helps to study the speed of market expectations formation and information diffusion

- Running the event study separately on two different samples helps to test economic hypotheses (in this case on principal-agent conflict) based on the prior that the economic hypothesis should apply more strongly to one group versus the other

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=961034

Case study 2: Share buybacks

Table 1
Descriptive Statistics on Open Market Share Repurchases

Univariate statistics for 3,481 open market share repurchase announcements between 1991 and 2001. CAR[-1,+1] is the cumulative abnormal announcement return over the three days around the repurchase announcement date using the market model with an equally weighted CRSP index. Fraction sought is the fraction of shares that the company announced it wishes to repurchase. Prior six-month raw returns is the cumulative return of the company in the six months prior to the repurchase announcement.

Year	Number of events	CAR[-1,+1]	Fraction sought	Prior six-month raw returns
1991	88	2.62%	6.90%	2.57%
1992	129	2.97%	8.10%	3.70%
1993	146	2.45%	6.99%	2.04%
1994	300	1.61%	6.32%	-1.50%
1995	299	2.21%	6.10%	3.71%
1996	407	2.21%	6.61%	3.78%
1997	394	2.40%	6.98%	8.11%
1998	682	2.13%	8.09%	-4.75%
1999	549	2.64%	8.35%	-3.13%
2000	302	2.66%	7.99%	-2.95%
2001	185	3.53%	7.16%	5.12%
All years	3481	2.39%	7.37%	0.43%

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

1. Urs Peyer, Theo Vermaelen, **The Nature and Persistence of Buyback Anomalies**, *The Review of Financial Studies*, Volume 22, Issue 4, April 2009, Pages 1693–1745

2. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

3. Despite abnormally positive returns following stock buyback announcements being known since the late 1980s (e.g. Lakonishok and Vermaelen '90), this anomaly has persisted in both short term market reaction ***and also long term multi-month abnormal positive returns following the announcement***

4. Table on the left shows short term [-1, +1], 3 day event window cumulative abnormal stock price reaction

Case study 2: Long term abnormal return after stock buyback announcements

Panel A and B: Full Sample and Subsamples by Book-to-Market Quintiles

Panel A: Fama-French IRATS

Months	Full Sample		BM lowest (glamour stocks)		BM 2		BM 3		BM 4		BM highest (value stocks)	
	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic
(+1,+12)	2.67%	3.077**	1.19%	0.403	4.90%	2.519*	5.00%	2.820**	2.39%	1.557	5.80%	1.941$
(+1,+24)	10.54%	8.015***	7.18%	1.671$	10.85%	3.544***	12.70%	4.598***	7.69%	3.071**	20.63%	4.847***
(+1,+36)	18.60%	11.328***	11.00%	2.027*	14.44%	3.570***	18.84%	5.061***	17.36%	5.131***	29.11%	5.553***
(+1,+48)	24.25%	12.529***	14.87%	2.136*	17.08%	3.353***	20.42%	4.425***	19.71%	4.536***	28.89%	4.473***
Obs	3481		439		699		820		900		623	

Panel B: Fama-French Calendar-Time Approach

	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic
12 mths	0.52%	3.11**	0.40%	1.08	0.19%	0.77	0.61%	2.74*	0.60%	3.30**	0.58%	1.71
24 mths	0.50%	3.77***	0.41%	1.36	0.37%	1.29	0.51%	2.39**	0.57%	2.97**	0.90%	3.04**
36 mths	0.45%	3.52***	0.41%	1.53	0.11%	0.56	0.48%	2.93**	0.65%	3.32***	0.84%	3.49***
48 mths	0.44%	3.56***	0.41%	1.62	0.10%	0.54	0.43%	2.70**	0.57%	3.76***	0.83%	3.74***

Panel C and D: Subsamples by Firm Size Quintiles

Panel C: Fama-French IRATS

Months around repurchase announcement	Largest firm quintile		Second largest firm quintile		Middle firm quintile		Second smallest firm quintile		Smallest firm quintile	
	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic
(+1,+12)	2.61%	2.061*	4.01%	2.447*	2.52%	1.389	2.92%	1.036	19.60%	2.444*
(+1,+24)	7.21%	3.995***	8.53%	3.499***	13.57%	4.588***	20.20%	4.820***	27.80%	2.632**
(+1,+36)	10.95%	4.788***	15.10%	4.900***	24.60%	6.670***	34.17%	6.593***	47.06%	3.283**
(+1,+48)	12.91%	4.780***	18.96%	5.183***	33.92%	7.688***	40.84%	6.883***	54.66%	3.679***
observations	992		870		830		620		169	

Panel D: Fama-French Calendar-Time Approach

	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic
12 months	0.48%	2.29*	0.25%	0.97	0.49%	1.69	0.53%	1.68	1.08%	1.58
24 months	0.35%	2.22$	0.49%	2.21$	0.60%	2.64*	0.77%	2.83**	0.98%	1.74
36 months	0.36%	2.49*	0.28%	1.59	0.52%	2.34*	0.73%	2.91**	1.35%	1.78
48 months	0.28%	2.01$	0.24%	1.43	0.58%	2.80*	0.67%	2.87**	1.38%	2.28*

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

1. Departing significantly from the daily event windows used so far in case study 1, this table shows long term cumulative abnormal announcements (CAR) following buyback announcements for 12, 24, 36 and 48 **months** after the announcement

2. In the table on the left, authors also segment their CAR analysis **according to 'book to market' quintiles, and also 'firm size'**

3. We can consider 'book to market' quintiles as a measure of the 'value' factor. **Together with 'firm size', they make up the other 2 'factors' in Fama-French 3 factor model apart from market beta**

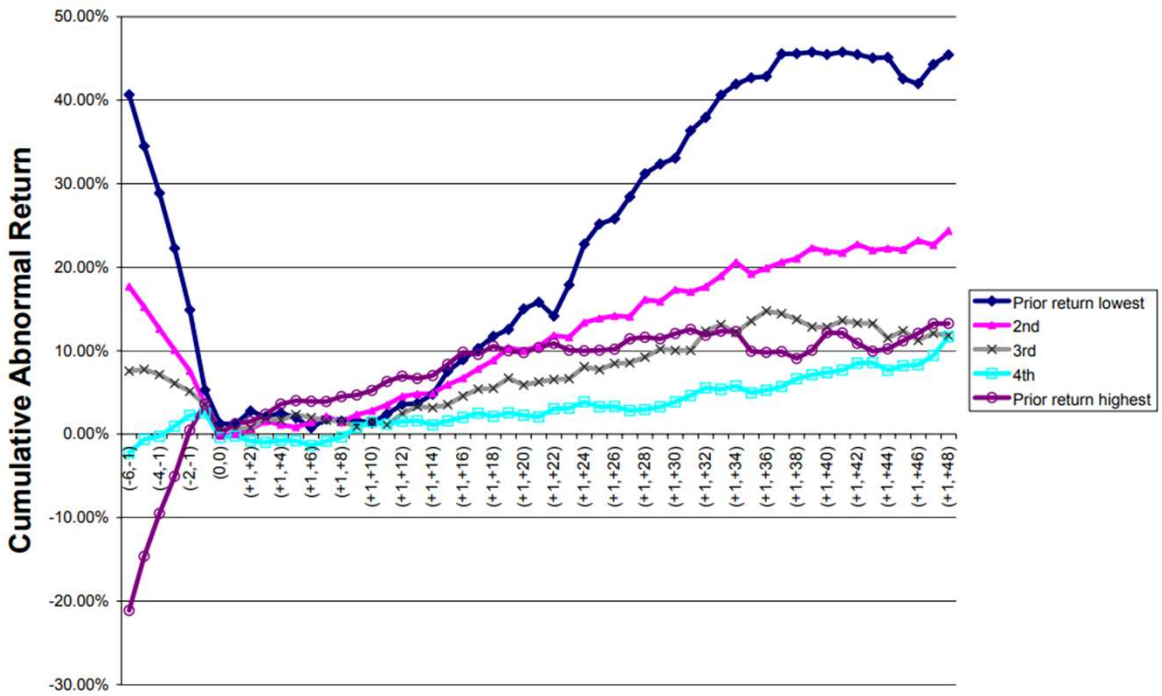
4. **Why did the authors further segment sample by FF3 factor quintiles** for this table?

5. **Why was this not done (and unnecessary) for [-1,+1], 3 day window on previous slide?**

Case study 2: Cumulative Long Run Abnormal Returns

Figure 1
Prior Return and Long-Run Abnormal Returns

The figure presents cumulative abnormal returns based on the Fama-French three-factor model and IRATS. For a description of the methodology see Table 3. The portfolios (quintiles) presented are formed based on the raw returns of stocks in the six months prior to the open market share repurchase announcement relative to the distribution of all CRSP firms' stock returns over the same period.



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Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

1. Long term (up to 48 month) CAR following buyback announcement illustrate return persistence, particularly for stocks that were worst performing prior to the buyback announcement

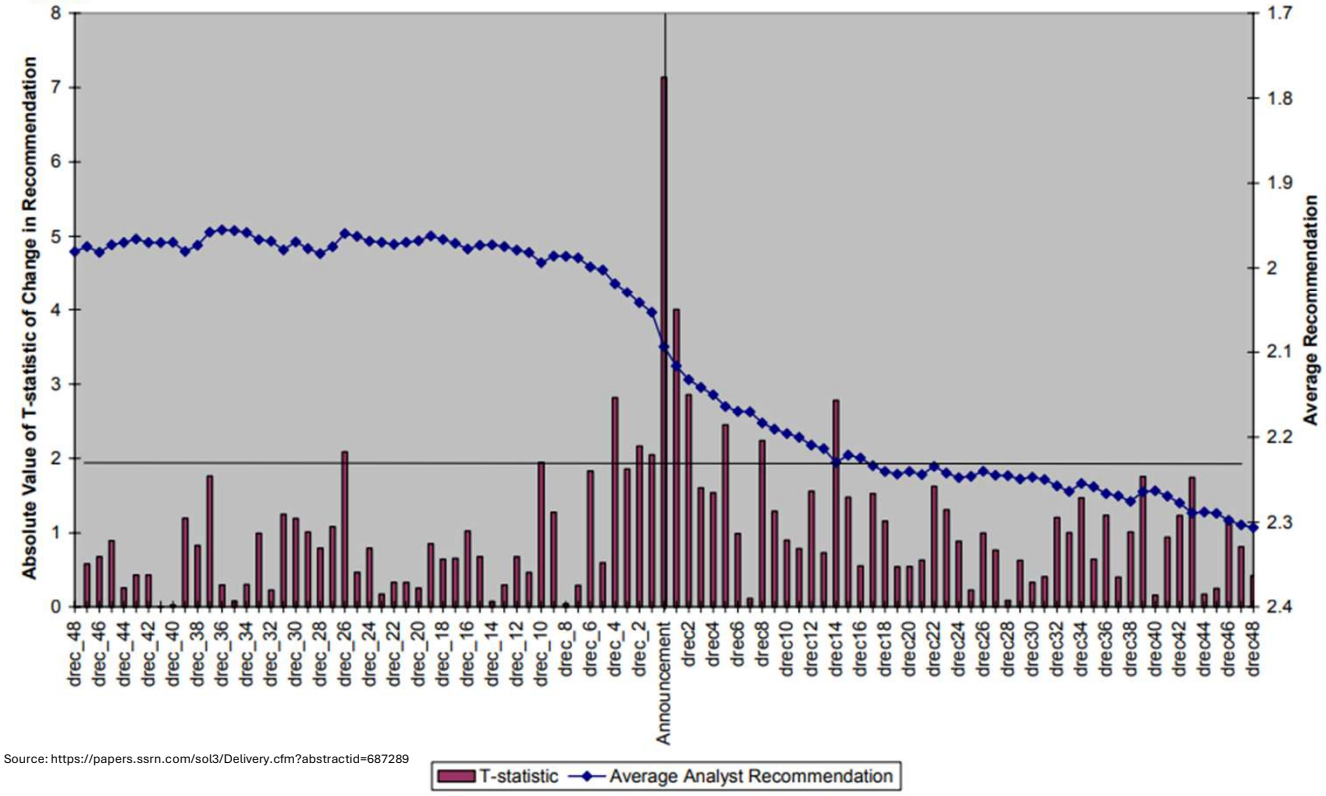
2. Given the time horizon and persistence, do you think this event study is consistent with:
   a. Informational signalling
   b. Public exchange supply-demand dynamics induced by the buyback
   c. (choose option which explains greatest fraction of CAR)

3. As background, stock buyback resolutions are typically board resolutions that authorize the purchase of "up to" a number of, or dollar value equivalent of shares prior to a specific date. Management may be given some flexibility in how to execute the buyback (for instance, in terms of how many shares to 'buy back' on any given day)

Case study 2: Market Sentiment following open market buyback announcement

Figure 6
Analyst Recommendations for Stocks that Announce an Open Market Buyback

The figure displays the average recommendation for a given month relative to the month in which the firm announced an open market share repurchase. We report average recommendations starting 48 months prior (indicated with drec_) to the announcement until 48 months after (indicated by drec). The recommendations are taken from IBES. Out of the 3481 events, 1836 stocks are covered by analysts who are recorded in IBES. The right scale indicates the average recommendation (1=strong buy; 2=buy; 3=hold; 4=sell; 5=strong sell). The left scale indicates the t-statistic for the change in the average analyst recommendation. The t-statistic is computed as the monthly change in average recommendation divided by the ratio of the standard deviation of the monthly change to the square root of the number of observations per event month. The standard deviation is based on the cross-sectional distribution of each event firm's change in average recommendation from t-1 to t (measured in months). A line is drawn at the level of 2 for the t-statistic.



Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

1. Stock buybacks in event time appear to be a reaction by company management to market undervaluation (including by sell side analysts)

2. Immediately prior to the buyback announcement, there appears to be a sharp / almost discontinuous change in market sentiment regarding the firm

3. This suggests a possible mode of enhancing any trading strategy conditioned on 'trend following' analyst recommendations

4. Firms which have a sharp decrease in analyst recommendations **AND** which do not have sufficient 'cash on hand' to conduct buybacks may be a better downside trend following candidate

Case study 2: Long term abnormal return after stock buyback announcements (robustness test on market model with liquidity factor)

Panel A: Fama-French IRATS including Pastor and Stambaugh (2003) value-weighted liquidity factor

Months	Full Sample Without liquidity factor		Full Sample With liquidity factor		Smallest firm quintile		BM highest (value stocks)	
	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic
(+1,+12)	2.67%	3.077**	3.41%	3.922***	17.33%	2.649**	1.88%	0.695
(+1,+24)	10.54%	8.015***	11.61%	8.817***	21.62%	2.701**	13.82%	3.524***
(+1,+36)	18.60%	11.328***	19.69%	11.980***	36.82%	3.252**	23.58%	5.039***
(+1,+48)	24.25%	12.529***	25.67%	13.240***	46.02%	3.842***	30.19%	5.596***
obs	3481		3481		169		623	

Panel B: Fama-French Calendar-Time Approach including the Pastor and Stambaugh (2003) value-weighted liquidity factor

	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic
12 mths	0.52%	3.11**	0.56%	3.85***	1.09%	1.56	0.62%	1.81
24 mths	0.50%	3.77***	0.55%	4.66***	1.00%	1.74	0.94%	3.67***
36 mths	0.45%	3.52***	0.48%	4.15***	1.17%	1.78	0.87%	3.84***
48 mths	0.44%	3.56***	0.47%	4.14***	1.21%	2.28**	0.86%	4.10***

1. In this slide, the authors add to their segmentation by the FF3 factors by also testing if observed CAR is different conditioning on a measure of the stock's market liquidity

2. This is because, for stocks with low market liquidity, the buyback offer might provide exit liquidity for existing shareholders (around a fixed price) and therefore reduce the liquidity risk premium associated with holding the stock

3. From a purely methodology standpoint, is this robustness test necessary for the case of a short term [-1,+1] 3 day window CAR?

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

Case study 2: Long term abnormal return after stock buyback announcements (robustness test on market model with price momentum / selection proxy)

Panel A: Fama-French IRATS

Months around repurchase announcement	Prior return lowest		Prior return 2		Prior return 3		Prior return 4		Prior return highest	
	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic
(-6,-1)	-40.65%	-29.927***	-17.68%	-18.056***	-7.55%	-8.621***	2.26%	2.529*	21.12%	16.717***
(-5,-1)	-34.50%	-27.600***	-15.22%	-16.891***	-7.75%	-9.911***	0.64%	0.807	14.64%	13.050***
(-4,-1)	-28.88%	-25.886***	-12.60%	-15.674***	-7.14%	-10.167***	0.26%	0.37	9.51%	9.629***
(-3,-1)	-22.27%	-22.098***	-10.07%	-14.250***	-6.08%	-9.809***	-0.95%	-1.506	5.05%	6.111***
(-2,-1)	-14.87%	-18.037***	-7.57%	-12.765***	-5.13%	-9.687***	-2.27%	-4.435***	-0.48%	-0.752
(-1,-1)	-5.29%	-8.357***	-3.63%	-7.902***	-3.54%	-8.111***	-2.52%	-6.508***	-3.60%	-7.638***
(0,0)	1.24%	1.277	-0.19%	-0.34	0.54%	1.109	-0.40%	-0.912	0.11%	0.211
(+1,+12)	3.55%	1.195	4.51%	2.286*	2.48%	1.479	1.54%	0.962	6.93%	4.025***
(+1,+24)	22.75%	5.200***	13.36%	4.459***	8.07%	3.024**	3.85%	1.561	9.97%	3.563***
(+1,+36)	42.85%	7.534***	19.88%	4.982***	14.76%	4.234***	5.26%	1.589	9.80%	2.735**
(+1,+48)	45.44%	6.192***	24.36%	4.843***	11.81%	2.647**	11.64%	2.758**	13.24%	2.761**
observations	740		668		650		664		759	

Panel B: Fama-French Calendar-Time Approach

	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic
12 months	0.01%	0.22	0.08%	0.31	0.21%	0.69	0.14%	0.45	0.60%	2.80*
24 months	0.55%	1.91$	0.47%	2.38*	0.33%	1.16	0.17%	0.87	0.41%	2.05$
36 months	0.96%	2.24*	0.52%	3.16**	0.35%	1.40	0.21%	1.31	0.25%	1.49
48 months	0.68%	2.55*	0.48%	3.07**	0.29%	1.28	0.25%	1.53	0.22%	1.42

Panel C: Fama-French Three-factor Model plus Momentum Factor of Carhart, IRATS

Months around repurchase announcement	Prior return lowest (stratified by year)		Prior return 2		Prior return 3		Prior return 4		Prior return highest	
	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic	CAR	t-statistic
(-6,-1)	-38.13%	-25.532***	-15.75%	-14.676***	-6.45%	-6.627***	2.31%	2.365*	20.79%	15.300***
(-5,-1)	-31.47%	-23.132***	-13.31%	-13.448***	-6.56%	-7.494***	0.57%	0.656	14.11%	11.659***
(-4,-1)	-26.35%	-21.920***	-10.82%	-12.224***	-6.00%	-7.546***	0.10%	0.127	8.96%	8.334***
(-3,-1)	-19.70%	-18.385***	-8.32%	-10.688***	-4.91%	-6.937***	-1.16%	-1.660$	4.42%	4.873***
(-2,-1)	-13.19%	-15.107***	-6.57%	-10.164***	-4.14%	-6.730***	-2.68%	-4.641***	-1.13%	-1.615
(-1,-1)	-4.56%	-6.775***	-3.23%	-6.601***	-2.90%	-5.624***	-2.87%	-6.529***	-4.28%	-8.243***
(0,0)	2.44%	2.464*	0.36%	0.605	0.90%	1.691$	-0.62%	-1.267	-0.86%	-1.533
(+1,+12)	9.51%	3.053**	7.73%	3.741***	5.27%	2.924**	5.73%	3.413***	10.10%	5.495***
(+1,+24)	32.01%	7.206***	19.37%	6.321***	13.29%	4.770***	11.14%	4.481***	17.60%	6.121***
(+1,+36)	50.37%	9.359***	31.13%	8.152***	25.01%	6.689***	16.80%	5.295***	20.97%	5.937***
(+1,+48)	60.69%	10.007***	42.68%	9.527***	30.34%	7.022***	24.71%	6.487***	27.07%	6.210***
observations	740		668		650		664		759	

Panel D: Fama-French Three-factor Model plus Momentum Factor of Carhart, Calendar-Time Approach

	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic	Monthly Average AR	t-statistic
12 months	0.40%	1.22	0.27%	1.05	0.88%	2.41*	0.73%	3.37**	0.98%	3.83***
24 months	0.94%	3.48***	0.65%	3.19**	0.61%	2.16$	0.58%	2.92**	0.80%	4.48***
36 months	1.02%	4.45***	0.66%	3.88***	0.73%	2.09*	0.58%	3.24**	0.53%	3.80***
48 months	0.99%	5.28***	0.65%	4.09***	0.72%	2.69*	0.54%	3.38**	0.51%	3.60***

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

1. Similar to the spirit of previous slides, the authors test if abnormal returns following a stock buyback announcement is significantly different conditional on prior stock price momentum for the stock

2. Why?

3. It is possible that the long term positive abnormal CAR is merely a continuation of prior price momentum (i.e. investor underreaction to a prior catalyst) rather than the buyback

4. Similar to before, this robustness test is not necessary for short term event window studies

Case study 2: Profits for Trading Tender Offer Expirations

Reported are cumulative abnormal returns following the strategy of buying shares six days prior to the first expiration date, tendering the shares and selling those shares not repurchased by the company on the market 12 (2) days after the final expiration date. The strategy is only executed if the price six days prior to the first expiration date is at least 3% below the repurchase tender price. This results in 80 events. Panel A calculates the returns using the marginal pricing rule where Return = $[F_P/F_T \times P_T + (1- F_P/ F_T) \times P_{12}] / P_{-6} -1$. F_P (F_T) is the number of shares repurchased (tendered) relative to the number of shares outstanding prior to the repurchase. P_T, P_{12} (P_{-6}) are the tender offer price and the stock price 12 days after (6 days prior to) the final (first) expiration date. We report the cumulative abnormal returns by subtracting the market return during the period from the strategy's return. Panels B and C report expected returns if the market priced the stock according to an average pricing rule E(Return) = $[F_P \times P_T + (1- F_P) \times P_{12}] / P_{-6} -1$, i.e., where the market assumes all shares are tendered (with $F_T = 1$). ***, ** indicate significance levels of 1 and 5 percent, respectively. Obs is Observations.

Panel A: Cumulative Abnormal Returns Based on Marginal Pricing Rule

Sample Period (observations)	Mean	t-value	Median	% Positive
1987-2001 (80 obs)	8.6%	5.5	4.1%***	84%
1987-1995 (51 obs)	8.2%	5.3	5.1%***	87%
1996-2001 (29 obs)	9.3%	2.8	3.0%**	79%

Panel B: Cumulative Abnormal Returns Based on Average Pricing Rule. Selling 12 days after the final expiration date

Sample Period (observations)	Mean	t-value	Median	% Positive
1987-2001 (80 obs)	1.54%	1.52	1.27%	58%
1987-1995 (51 obs)	2.47%	2.26	2.73%**	62%
1996-2001 (29 obs)	0.00%	0.00	-0.004%	48%

Panel C: Cumulative Abnormal Returns Based on Average Pricing Rule. Selling two days after the final expiration date

Sample Period (observations)	Mean	t-value	Median	% Positive
1987-2001 (80 obs)	1.67%	2.64	1.25%**	64%
1987-1995 (51 obs)	2.69%	3.44	2.23%**	74%
1996-2001 (29 obs)	0.03%	0.04	-0.005%	47%

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

Summarizing case study 2

1. Long horizon event studies require (at a minimum) significant robustness testing to ensure that results on 'abnormal returns' are robust to choice of 'market benchmark'
   a. Note that the authors choose to test robustness include not just FF3 factors, but also a variety of other less well known factors, e.g. Carhart momentum factor, liquidity factor etc
   b. As of 2025, there are too many factors extant to conduct comprehensive robustness testing against all factors
   c. Choice of factors to conduct robustness testing against should therefore be motivated by economic intuition / hypothesis on whether a specific factor is plausibly correlated to the event being studied.
   d. For example, stock return momentum (whether positive or negative) prior to a buyback announcement may plausible have a causal relationship in terms of motivating firm management to decide on having a buyback

2. Similar to the case of seasoned equity offering, we need to consider endogeneity of corporate policies for internally initiated events

Source: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=687289

Joint hypothesis test problem and event studies [1/2]

1. Many tests of market efficiencies in finance are subject to the "joint hypothesis test" problem

2. Consider a simple linear estimation of policy effect (policy variable is dichotomous 1 or 0):
   a. $R_{it} = a_i + b_1R_{mt} + b_2D_{it} + e_{it}$
   b. Here, D_{it} may denote whether the stock has a buyback program ongoing (1 if so else 0), whether the stock is undergoing a secondary equity offering (1, else 0), etc
   c. In this estimation, we may wish to conclude that D_{it} results in some persistent deviation of security i return from that which is predicted by CAPM, and therefore the market is inefficient
   d. But we are essentially testing 2 hypotheses simultaneously in this estimation
   e. First, estimated coefficient on D_{it} is statistically significant and signed in line with economic intuition
   f. Second, that the model of what is 'rational returns' as denoted by CAPM is *correct*

3. Unfortunately, it is by no means straightforward what is the 'appropriate' market model to 'benchmark' against
   a. Should it be CAPM?
   b. Why not the Fama-French 3 factor model?
   c. Why not the Carhart 4 factor model?
   d. … …

4. If we wish to demonstrate that a particular variable / event is associated with abnormal returns different from that predicted by the market model, then any analysis of "what is abnormal returns" following or surrounding the event needs to be robust to choice of benchmark. This is by no means a trivial undertaking

Joint hypothesis test problem and event studies [2/2]

1. Severity of 'joint hypothesis test' problem depends profoundly on short / long horizon nature of the event study

2. Short horizon event studies involving the computation of abnormal returns over (say) several days **only (e.g -2 to +2 window, etc)** avoid difficulties of the joint hypothesis test problem.

    a. Over a sufficiently short horizon, we may plausibly assume that returns of the instrument due to 'market beta' is 0 (regardless of market model; CAPM, FF3, C4, etc). Hence, all 'returns' to studied instruments, if statistically significant, is assumed to accrue because of the event

    b. Consequently, short horizon event studies *avoid* the need to condition on a market model, although they do not *solve* the theoretical problem of an unobserved market model

    c. Short-horizon tests represent the "cleanest evidence we have on efficiency" (Fama, 1991, p.1602)

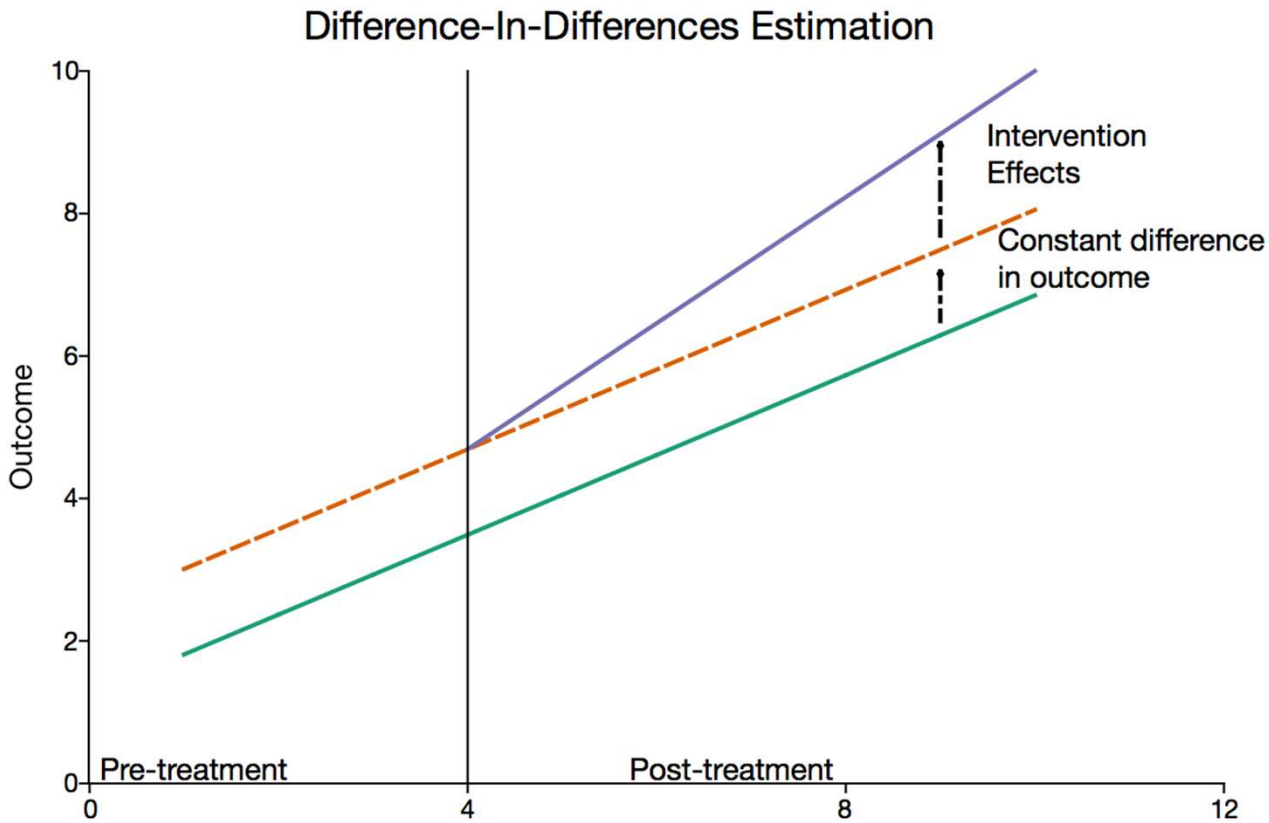
3. On the other hand, long horizon event studies are significantly more vulnerable to the joint hypothesis test and (at a minimum) likely require robustness testing across multiple market benchmark models

Short versus Long Run Event Studies

1. Effect of unobservable factors is likely to be more 'marked' in long run event studies versus short run studies
   a. For a short enough event window, we assume that impact of unobservable factors is likely to be 0
   b. For a long run event study, given any differences in post treatment performance, how do we know 'treatment group' performance was not due to (one or more) of the hundreds of asset pricing factors that are currently known?
   c. "Before – During – After" event studies are naturally less vulnerable to this objection compared to "Before – After" event studies

2. Alternatives to Long Run Event Studies for 'Before-After' event studies:
   a. RDD
   b. Difference in difference (commonly referred to as 'DID' in the literature) does not assume that both groups have the same 'factor exposures', but it assumes that differential performance 'since' the event is due to the event.

Difference in difference (DID) methodology



1. Chart on the left shows commonly used DID methodology. If we are interested in treatment effects on asset pricing:

2. **Should y axis be asset prices or asset returns**, if we wish to credibly abstract away from unobservable factor loading differences over medium horizons?

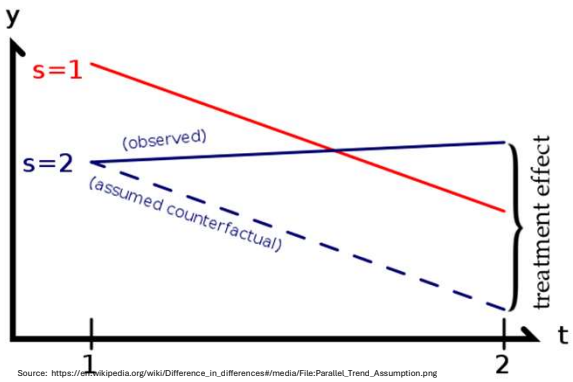
3. How would we estimate this in a regression format (OLS / Ridge / Huber / Lasso / etc)?

4. LHS variable is ____

5. Key variable of interest on the RHS is a ____ variable

DID methodology: Parallel trend assumption

1. DID requires a parallel trend assumption, which assumes that difference between both treatment and control groups are constant across horizon of the treatment, except for the 'event' or treatment.
    a. This assumption is valid if unobservable differences in factor loadings between both groups only have a (i) linear effect on asset returns, **and (ii)** factor loadings remain constant **and also** (iii) factor returns remain constant
    b. Assumption on factor 'linear effect on asset returns' is an implicit assumption of most asset pricing models formulated in the arbitrage pricing theory (APT) framework. It is *probably* true for any study conducted at decent scale (assets of sufficient tradable volume over significant periods of time) because faster than linear impact on returns are unlikely to be sustainable while slower than linear impact on returns are unlikely to result in statistically significant results in a linear model
    c. What is a prominent regularity in factor behaviour that could contradict above assumptions, and what are potential solutions? The regularity is _____
2. In practice, we can match both treatment and control groups on a selected number of observable covariates (e.g. market beta, firm size, industry and other 'very obvious' metrics) to obtain better / more credible results.
    a. It will not be possible to match on all the various hundreds of factor loadings, and the purpose of DID is to control for this. However, to the extent that we can control for some of the more significant factors, there is 'no loss' in doing so



Source: https://en.wikipedia.org/wiki/Difference_in_differences#/media/File:Parallel_Trend_Assumption.png

Case study: Differential impact of firm size on sensitivity to investor sentiment

1. In our previous class, we used PLS and PCA to form measures of investor attention.

2. In today's case study, we look at measures of investor *sentiment*
   a. Timothy K. Chue, Ferdinand A. Gul, G. Mujtaba Mian, Aggregate investor sentiment and stock return synchronicity, Journal of Banking & Finance, Volume 108, 2019, 105628, ISSN 0378-4266
   b. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=3453023

3. Caution:
   a. As mentioned in class, attention, sentiment and relevance are 3 related, but conceptually distinct measures of impact of 'news/media content' on asset returns
   b. Investor sentiment, which we are discussing today can be both positive and negative
   c. On the other hand, 'investor attention' which we discussed last week is floored at 0 and cannot be negative
   d. A crude benchmark model can be: impact $\propto$ (sentiment $\times$ attention $\times$ relevance) where latter 2 variables are >=0 (as with all benchmark models, it merely forms a basis for further improvement)

How do we measure sentiment?

SENT	Sentiment index in Baker and Wurgler (2006); updated version of Eq. (2) in that paper; based on first principal component of FIVE (standardized) sentiment proxies
SENT⊥	Sentiment index in Baker and Wurgler (2006); updated version of Eq. (3) in that paper; based on first principal component of FIVE (standardized) sentiment proxies where each of the proxies has first been orthogonalized with respect to a set of six macroeconomic indicators
pdnd	Value-weighted dividend premium defined following Baker and Wurgler (2004) (values differ slightly from there due to subsequent improvements in the CRSP/Compustat merge procedure). The indexes use the t-12 value. (From SAS program divpremmonth_jaz_20210111.sas)
ripo	First-day returns on IPOs from Ibbotson, Sindelar, and Ritter (1994) and updates (NIPO-weighted average of monthly RIPOs) from Jay Ritter's website. We provide monthly data here; the indexes use the nipo-weighted average over the prior twelve months to smooth noise, and then use the t-12 value of the result
nipo	IPO volume from Ibbotson, Sindelar, and Ritter (1994) and updates from Jay Ritter's website. We provide monthly data here; the indexes use the sum of nipo over the prior twelve months to smooth noise
cefd	Closed-end fund discount from Neal and Wheatley (1998) for 1934 to 1964 ('domestic stock funds'); Lakonishok, Shleifer, Vishny (1991) for 1965 to 1985 (general equity funds only); CDA/Wiesenberger for 1986; Herzfeld from 1987-2010; Morningstar from 2011. (unlevered general equity only) (equal-weighted)
s	Equity share in new issues defined following Baker and Wurgler (2000), i.e., the total volume of equity issues over the prior twelve months divided by the total volume of equity and debt issues over the prior twelve months from Federal Reserve Bulletin
indpro	Industrial production index; we provide monthly data here; the orthogonalized index uses growth over the t-12 value
consdur	Nominal durables consumption; we provide monthly data here; the orthogonalized index uses growth in the real value over the t-12 real value
consnon	Nominal nondurables consumption; we provide monthly data here; the orthogonalized index uses growth in the real value over the t-12 real value
consserv	Nominal services consumption; we provide monthly data here; the orthogonalized index uses growth in the real value over the t-12 real value
recess	NBER recession indicator
employ	Employment; we provide monthly data here; the orthogonalized index uses growth over the t-12 value
cpi	Consumer price index

1. Download raw sentiment data here: https://pages.stern.nyu.edu/%7Ejwurgler/data/SENTIMENT.xlsx
2. Methodology:https://pages.stern.nyu.edu/%7Ejwurgler/papers/wurgler_baker_investor_sentiment.pdf
3. Personal critique: There is some clear overlap between sentiment and also attention measures. My suggestion for refinement is to use PCA or PLS (which we learnt last week!) to form orthogonal measures of sentiment and attention.

What is dependent variable for DID?

$$Synchronicity_{j,t} = \ln\left(\frac{R^2_{j,t}}{1-R^2_{jt}}\right)$$

1. Goal is to study how much equity returns are explained by the 'sentiment' factor

2. Coefficient of determination, R^2 is a measure of synchronicity between stock j's return and factor returns

3. This measure, however, is bounded between 0 and 1 and suffers from large skewness and kurtosis

4. To circumvent these issues, authors apply a logistic transformation to raw R^2 to obtain our measure of return synchronicity

5. $Synchronicity_{j,t}$, for stock j in month t

Summary statistics of sentiment on stock returns [1/2]

Table 5

Cross-sectional variation in the sentiment-synchronicity relation: Bivariate sorts

This table reports the cross-sectional variation in the relation between investor sentiment and stock return synchronicity. At the beginning of each calendar month, we sort all stocks based on a composite rank that captures the sensitivity of the stock to investor sentiment. The composite rank is based on the first principal component of the ranks of a stock on six stock characteristics, namely size, age, stock return volatility, dividend payout, the MB ratio and the stock price level. Stocks that fall in the two extreme quintiles are labelled as 'High Sensitivity' and 'Low Sensitivity', and the remaining 60% of stocks are classified as 'Medium'. Within each group, we then compute average synchronicity within a month. Months are then sorted into quintiles based on the value of the Baker-Wurgler investor sentiment index at the end of the previous month. Finally, we compute average synchronicity for each subgroup within a sentiment quintile by taking a simple average of all of the synchronicity measures for that subgroup that fall in that sentiment quintile. The row 'Difference (Low - Moderate)' reports the difference in stock return synchronicity between the 'Low' and 'Moderate' sentiment quintiles. The row 'Difference (High - Moderate)' reports the difference in stock return synchronicity between the 'High' and 'Moderate' sentiment quintiles. Panel A reports the analysis for the total sample. Panels B and C repeat the same analysis for subsamples of liquid and illiquid stocks, respectively. At the end of every month, all stocks are ranked based on their *PctZero* – the proportion of the stocks' zero-return days in a month. Stocks that are below the median *PctZero* for a given month are considered 'Liquid', and stocks that are above the median *PctZero* are considered 'Illiquid' The analysis is based on monthly data from October 1966 to January 2011. The *t*-statistics are reported in parentheses and are based on Newey-West (1987) standard errors.

Panel A: All stocks

	Average monthly synchronicity			Cross-sectional difference
Quintiles of sentiment	High sensitivity	Medium	Low sensitivity	
(1)	(2)	(3)	(4)	(5) = (2) - (4)
1 (Low)	-0.319	-0.169	0.070	
2	-0.351	-0.232	0.154	
3 (Moderate)	-0.349	-0.218	0.132	
4	-0.317	-0.173	0.210	
5 (High)	-0.228	-0.121	0.087	
Difference (Low - Moderate)	0.029	0.049	-0.062	0.092
t-statistics	(0.96)	(1.13)	(-1.05)	(1.85)
Difference (High - Moderate)	0.121	0.098	-0.045	0.166
t-statistics	(3.33)	(2.08)	(-0.78)	(3.20)

Summary statistics of sentiment on stock returns [2/2]

Table 5 (Continued)

Panel B: Liquid stocks

Quintiles of sentiment	Average monthly synchronicity: High sensitivity	Medium	Low sensitivity	Cross-sectional difference
(1)	(2)	(3)	(4)	(5) = (2) - (4)
1 (Low)	-0.235	-0.071	0.097	
2	-0.308	-0.106	0.198	
3 (Moderate)	-0.277	-0.107	0.174	
4	-0.238	-0.043	0.248	
5 (High)	-0.144	0.002	0.121	
Difference (Low - Moderate)	0.042	0.036	-0.077	0.119
t-statistics	(1.04)	(0.74)	(-1.33)	(2.53)
Difference (High - Moderate)	0.133	0.109	-0.052	0.185
t-statistics	(2.94)	(2.05)	(-0.92)	(3.98)

Panel C: Illiquid stocks

Quintiles of sentiment	Average monthly synchronicity: High sensitivity	Medium	Low sensitivity	Cross-sectional difference
(1)	(2)	(3)	(4)	(5) = (2) - (4)
1 (Low)	-0.345	-0.265	-0.095	
2	-0.368	-0.334	-0.041	
3 (Moderate)	-0.375	-0.308	-0.042	
4	-0.342	-0.288	0.016	
5 (High)	-0.256	-0.235	-0.104	
Difference (Low - Moderate)	0.030	0.043	-0.053	0.083
t-statistics	(1.07)	(1.26)	(-0.81)	(1.47)
Difference (High - Moderate)	0.119	0.073	-0.062	0.180
t-statistics	(3.60)	(1.86)	(-0.95)	(3.28)

DID estimation on differential sensitivity to investor sentiment based on firm size

Table 6
Cross-sectional variation in the sentiment-synchronicity relation: Multiple regressions

This table reports the results from estimating regressions of the following form:

$$Synchronicity_{j,t} = c_0 + c_1 SentResPos_{t-1} + c_2 AbsSentResNeg_{t-1} + c_3 SentResPos_{t-1} \times Rank6V_{j,t-1} + c_4 GDP_{t-1} + c_5 GDP_{t-1} \times Rank6V_{j,t-1} + c_6 Illiq_{t-1} + c_7 Illiq_{t-1} \times Rank6V_{j,t-1} + c_8 PeriodCount_t + c_9 PeriodCountSq_t + c_{10} ROESynchronicity_{j,t-1} + \varepsilon_{j,t}$$

	Dependent variable = *FF4F_Synchronicity*		
Explanatory variables	(1)	(2)	(3)
Intercept	-0.019	0.015	0.002
	(-0.28)	(0.22)	(0.03)
SentResPos	0.103	0.137	0.108
	(3.74)	(4.75)	(3.84)
AbsSentResNeg	0.002	0.007	-0.003
	(0.06)	(0.19)	(-0.08)
SentResPos x Rank6V	-0.010	-0.017	-0.010
	(-4.56)	(-6.65)	(-4.65)
PctZero (Firm-level, cross-sectionally demeaned)		0.001	
		(3.44)	
PctZero (Firm-level, cross-sectionally demeaned) x Rank6V		-0.001	
		(-23.76)	
Illiq (Firm-level, cross-sectionally demeaned)			0.0003
			(1.23)
Illiq (Firm-level, cross-sectionally demeaned) x Rank6V			-0.001
			(-9.37)
GDP	0.021	0.029	0.019
	(1.92)	(2.51)	(1.74)
*GDP*Rank6V*	-0.011	-0.013	-0.010
	(-6.95)	(-6.97)	(-6.38)
PeriodCount	-0.005	-0.005	-0.005
	(-11.13)	(-11.54)	(-11.26)
PeriodCountSqr	0.000	0.000	0.000
	(12.72)	(13.30)	(12.90)
Rank6V	0.071	0.061	0.065
	(30.75)	(24.40)	(28.91)
ROE Synchronicty	0.006	0.005	0.005
	(4.87)	(4.69)	(4.37)
Adj R^2	0.096	0.110	0.100
Number of pooled stock-month obs.	1,733,113	1,733,094	1,714,140

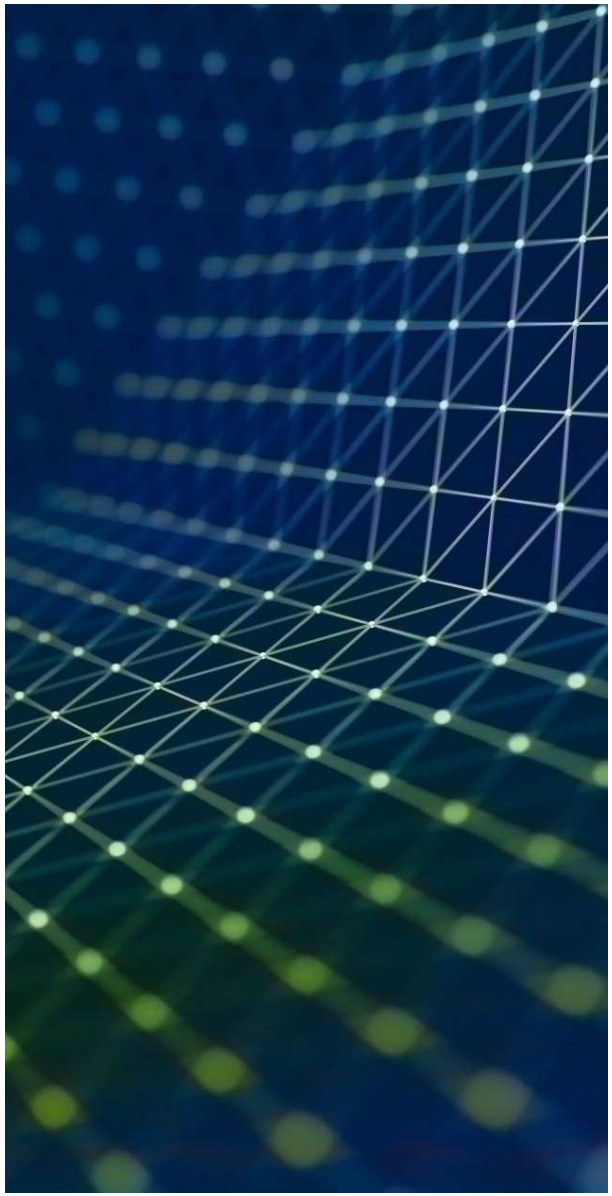
1. In this estimation, Rank6V is the composite rank based on the first principal component of the ranks of all stocks based on six characteristics: size, age, stock return volatility, dividend payout, MB ratio and the stock price level

2. Low values of Rank6V signify greater sensitivity to investor sentiment.

3. Main variable of interest in the estimation is interaction variable SentResPos x Rank6V

4. DID methodology is therefore based on measuring differences in 'treatment' **not just between not just 2 groups, but between 6 groups!!**

5. Nothing in DID methodology says parallel trends assumption needs to be restricted to 2 groups only; we can generalize to an arbitrary but finite number of groups



Rate models

Poisson models

Hazard rates

While event studies measure the impact of 'treatment' or 'exogeneous' events, how do we estimate probability of such events in the first place?

1. Event studies measure treatment effect conditional on the occurrences of endogenous treatment or exogeneous 'shock'

2. However, significant fraction of short term event studies are not easily tradable because they encompass [-x, +y] event windows, where –x denotes x days *before* event announcement / impact

3. We are not able to trade in advance, and only able to exploit [+1, +y] event reactions; in many cases, there may still exist significant reactions in that window, but asset price reactions are much reduced.
   a. This objection is less marked for longer term event studies, although (as discussed) these are vulnerable to joint hypothesis objection

4. If we are able to estimate conditional probability of such occurrences, then we can modify expected excess return:
   a. From [+1, +y] to …
   b. E(p)*[-x, +y] + (1-E(p))*[benchmark expected excess return, which is typically 0]

5. How do we form an estimate of E(p)?
   a. E(p) is similar to the *rate* of the event per unit time
   b. Difference is that probability refers to likelihood of event happening over a fixed period of time, while rate is # of events occurring / time unit. However, solving for one can significantly help is solving for the other

Rate models

1. In this segment, we introduce **two forms of rate models:**
2. **Poisson models** are used to model the rate of events which nevertheless can recur repeatedly. Examples:
    a. Market crashes
    b. Participating in M&A as an acquirer
    c. Number of patents generated per year
    d. Stock buybacks
    e. Rate of pollution by a firm
    f. Seasoned / secondary stock offerings
    g. Bond upgrades / downgrades

3. On the other hand, **hazard models** are used to model probability of rare events which can only occur once; following occurrence of the event, subject is permanently removed from the sample and no further information is obtainable on the subject. Examples:
    a. Participating in M&A as a target
    b. IPOs
    c. Firm bankruptcy (some firms can go bankrupt multiple times; airlines are notorious for this; nevertheless, we can view each iteration of the firm as different / distinct from previous generations – notwithstanding that they may have the same name. This assumption is further justified that each generation is likely to have different equity holders and management)
    d. Take privates (see above)

4. Permanent removal of subjects from the sample following event occurrence (in case of hazard models) try to **solve a bias problem that OLS is unable to address**
    a. For some (typically most…) subjects in the sample, the 'event' would not have been observed before end of the sample
    b. This biases sensitivity of covariates towards _____
    c. In a hazard model, we estimate conditional probability of an event *given* that event has not yet occurred

Poisson models and the Poisson distribution

1. Recall that the Poisson distribution is completely described by just 1 parameter

$$P(y_i = k) = \frac{\lambda_i^k}{k!} e^{-\lambda_i}$$

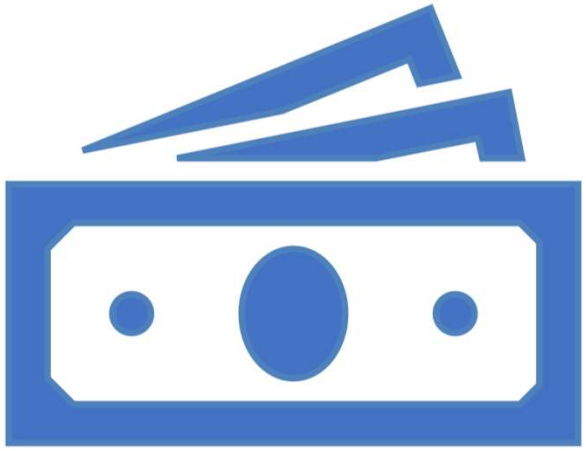
2. How do we estimate lambda given a dataset of count occurrences with a set of covariates?

3. Given dependent variable which is count of occurrences for each case and a vector of independent variables which we believe could affect λ for that case, we estimate λ as:

$$\log \lambda_i = \beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip}$$

4. As λ is constrained to always be non negative, we can fit this model via maximum-likelihood

Sanity checks on estimated λ



1. Since λ is both mean and standard deviation, one sanity check that we have estimated an appropriate value of λ is that around 95% of observations should fall within +-2(sqrt(λ)) of λ.

2. In practice, it is common for most datasets to exhibit 'fatter tails' and higher true dispersion.

3. If we estimate standard errors using th λ estimates, then our test statistics are likely to be overly inflated in many common cases, leading to overconfidence in model coefficients

Case study: Poisson estimation with fixed effects

1. Paper: Jonathan B. Cohn, Zack Liu, Malcolm I. Wardlaw, **Count (and count-like) data in finance**, Journal of Financial Economics, Volume 146, Issue 2, 2022, Pages 529-551, ISSN 0304-405X,

2. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=3800339

3. This paper replicates a number of studies in corporate finance where outcome variable is a rate (output per unit time)
   - Number of patents generated by a firm / year
   - Weight of pollution released by a facility in a given year

Table B1: Replication: Hirshleifer, Low, and Teoh (2012)

This table presents a series of regressions based on the regression specification in Table V column (3) of Hirshleifer, Low, and Teoh (2012). The unit of observation is a firm-year. The outcome variable is the number of patents a firm generates in a given year. The first column reproduces the results from the original paper, which estimates a log1plus regression. The next three columns present results from log1plus, log-linear, and Poisson regressions, based on our replication of the original data set. The final column presents results from log1plus regression where the sample is restricted to the sample usable in Poisson regression. Standard errors clustered at the firm level are presented below each coefficient. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively, based on a two-tailed t-test.

	Original Log1plus	Replication Log1plus	Replication Log-Linear	Replication Poisson	Replication Log1plus Poisson Sample
Overconfident CEO	0.111**	0.145**	0.288***	0.492***	0.155**
	(0.053)	(0.065)	(0.091)	(0.183)	(0.066)
Log(sales)	0.640***	0.402***	0.518***	0.818***	0.406***
	(0.051)	(0.038)	(0.042)	(0.072)	(0.038)
Log(PPE/Emp)	0.218***	0.211***	0.321***	0.688***	0.218***
	(0.051)	(0.053)	(0.083)	(0.175)	(0.054)
Stock return	0.052***	0.085***	0.060	0.175***	0.086***
	(0.012)	(0.023)	(0.037)	(0.058)	(0.023)
Institutional holdings	-0.113***	-0.503***	-0.467**	-0.855**	-0.518***
	(0.030)	(0.177)	(0.237)	(0.362)	(0.179)
Log(1+tenure)	-0.051**	-0.058*	-0.135***	0.044	-0.062*
	(0.025)	(0.035)	(0.046)	(0.080)	(0.035)
Log(1+delta)	0.014	0.034	0.110**	-0.075	0.034
	(0.036)	(0.031)	(0.044)	(0.092)	(0.031)
Log(1+vega)	0.218***	0.161***	0.208***	0.284***	0.165***
	(0.039)	(0.034)	(0.046)	(0.092)	(0.034)
Observations	8,939	6,482	3,121	6,326	6,326
Adjusted R2	0.507	0.483	0.516		0.479

Hazard models and survival analysis

Consider the hazard function, which models the probability of failure at time t, conditional upon survival up until time t

$$h_i(t) \approx \frac{P\left(t < T_i < t + \Delta t \mid T_i > t\right)}{\Delta t}$$

Unlike Poisson model, subjects in survival model **can only ever fail once.** It is used to model "non-repairable" items

There will also be right censoring in data, which is presence of subjects in dataset that have not experienced failure by end of sample period

What does the hazard function measure?

$$h_i(t) = h_0(t) \cdot \exp\left\{\beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip}\right\}$$

1. LHS measures the instantaneous rate of failure of the subject at time t

2. Specifically, it is the conditional failure rate, conditional on a subject having survived until time t

3. Units is rate of failure

Decomposing the hazard function

$$h_i(t) = h_0(t) \cdot \exp\left\{\beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip}\right\}$$

The hazard function is semi-parametric because:

1. The rate function $h_0(t)$ which is the baseline hazard function can be nonparametric
2. The exp{..} term is parametric, and it raises entire baseline function up or down by a uniform amount across all t
3. This is known as the proportional hazards assumption

Interpreting impact of a 1 unit increase in a covariate

What is impact of a 1 unit increase in x1?

$$\frac{h_i(t)}{h_j(t)} = \frac{h_0(t)\exp\{\beta_0 + \beta_1 \boxed{(x^\star + 1)} + \beta_2 x_2 + \cdots \beta_p x_p\}}{h_0(t)\exp\{\beta_0 + \beta_1 x^\star + \beta_2 x_2 + \cdots \beta_p x_p\}}$$

$$= \exp\{\beta_1(x^\star + 1 - x^\star)\}$$

$$= \exp(\beta_1).$$

Increase in hazard rate of $e^{\beta 1}$ is assumed to hold across all values of t

```
210 #HAZARD models
211 survival_df = pd.read_csv("survivaldata.csv")
212 survival_df.dropna(subset=['isdelisted'], inplace=True) #drop data where delisted status is unsure
213 survival_df.loc[survival_df.isdelisted=='N', 'dead'] = 0    #code a 1/0 variable based on whether delisted or not.  Python ne
214 survival_df.loc[survival_df.isdelisted=='Y', 'dead'] = 1
215 sizedummies = pd.get_dummies(survival_df['scalerevenue'])    #estimate probabiltiy of delisting based on revenue category
216 survival_df = pd.concat([survival_df, sizedummies], axis=1)
217 import statsmodels.formula.api as smf
218 survival_df.rename(columns = {'1 - Nano':'Nano','2 - Micro': 'Micro', '3 - Small': 'Small', '4 - Mid': 'Mid', '5 - Large': 'L
219 mod = smf.phreg("duration ~ 0 + Nano + Micro + Small + Mid + Large + Mega", survival_df, status = survival_df["dead"].values,
220 rslt = mod.fit()
221 rslt.summary()
222 survival_df['sizecat'] = 0
223 survival_df.loc[survival_df['Micro']==1, 'sizecat'] = 1
224 survival_df.loc[survival_df['Small']==1, 'sizecat'] = 2
225 survival_df.loc[survival_df['Mid']==1, 'sizecat'] = 3
226 survival_df.loc[survival_df['Large']==1, 'sizecat'] = 4
227 survival_df.loc[survival_df['Mega']==1, 'sizecat'] = 5
228 survival_df.sizecat
229 mod = smf.phreg("duration ~ 0 + sizecat", survival_df, status = survival_df["dead"].values, ties = "efron")
230 rslt = mod.fit()
231 rslt.summary()
```

Estimating hazard of firm delisting

- Lines 211 to 216: Import firm survival / delisting csv file and code up numerical 1/0 survival variables
- Lines 217 to 221: Run hazard model of firm delisting based on discrete revenue category dummies [why is this not optimal?]
- Lines 218 to 231: Run hazard model of firm delisting based on single size categorical variable

One weakness of this way of estimating is that each HR is relative to “all other firms”. For e.g. hazard ratio on “Small” is relative to “Nano”, “Micro” but also “Mid”, “Large”, and “Mega”. So it is not straightforward to interpret

```
<class 'statsmodels.iolib.summary2.Summary'>
"""
                      Results: PHReg
============================================================
Model:                    PH Reg       Sample size:     37850
Dependent variable:       duration     Num. events:     22114
Ties:                     Efron
------------------------------------------------------------
       log HR log HR SE   HR      t     P>|t|  [0.025 0.975]
------------------------------------------------------------
Nano   0.2059    0.0219 1.2286   9.4155 0.0000 1.1770 1.2824
Micro  0.0195    0.0211 1.0197   0.9236 0.3557 0.9784 1.0627
Small -0.3306    0.0222 0.7185 -14.8766 0.0000 0.6879 0.7505
Mid   -0.9237    0.0306 0.3970 -30.2113 0.0000 0.3739 0.4216
Large -1.7264    0.0517 0.1779 -33.3990 0.0000 0.1608 0.1969
Mega  -2.9883    0.5004 0.0504  -5.9725 0.0000 0.0189 0.1343
============================================================
Confidence intervals are for the hazard ratios
"""
```

```
In [4]: survival_df['sizecat'] = 0
   ...: survival_df.loc[survival_df['Micro']==1, 'sizecat'] = 1
   ...: survival_df.loc[survival_df['Small']==1, 'sizecat'] = 2
   ...: survival_df.loc[survival_df['Mid']==1, 'sizecat'] = 3
   ...: survival_df.loc[survival_df['Large']==1, 'sizecat'] = 4
   ...: survival_df.loc[survival_df['Mega']==1, 'sizecat'] = 5
   ...: survival_df.sizecat
   ...: mod = smf.phreg("duration ~ 0 + sizecat", survival_df, sta
   ...: rslt = mod.fit()
   ...: rslt.summary()
D:\Anaconda\lib\site-packages\statsmodels\duration\hazard_regressi
  warnings.warn("PHReg formulas should not include any '0' or '1'
Out[4]:
<class 'statsmodels.iolib.summary2.Summary'>
"""
                        Results: PHReg
================================================================
Model:                   PH Reg        Sample size:      37850
Dependent variable:      duration      Num. events:      22114
Ties:                    Efron
----------------------------------------------------------------
         log HR log HR SE   HR      t     P>|t|  [0.025 0.975]
----------------------------------------------------------------
sizecat -0.3376    0.0061 0.7135 -55.3201 0.0000 0.7050 0.7220
================================================================
Confidence intervals are for the hazard ratios
"""
```